



(b) (6) **ANALYTICAL METHOD TEST MEMO**

To: Administrative File STN 125752/0

From: (b) (6) CBER/ (b) (6)

Through: (b) (6), CBER/ (b) (6)

(b) (6) CBER/ (b) (6)

(b) (6), CBER/ (b) (6)

Product: SPIKEVAX

Applicant: ModernaTX, Inc.

Subject: Test results of pH and lipid nanoparticle size and polydispersity index for drug product of SPIKEVAX

1. pH determination

The test was performed per (b) (6) Doc000482 "pH Determination" on December 10, 2021. pH values of three DP sample lots were measured in duplicate with a pH standard of (b) (4) as the assay control. The result of the control was (b) (4), which met the acceptance criterion of 7.00 ± 0.05 .

The test results by (b) (6) and those from the firm are summarized in Table 1.

Table 1. Results of pH value

Lot	CBER pH	Firm pH
031H21A	(b) (4) (b) (4) Average: (b) (4)	(b) (4)
064H21A	(b) (4) (b) (4) Average: (b) (4)	(b) (4)
940875	(b) (4) (b) (4) Average: (b) (4)	(b) (4)

CBER's test results from all three DP sample lots were (b) (4) within the proposed pH specification of (b) (4), and met the acceptance criterion of the difference between duplicate measurements of (b) (4) pH unit.

2. Lipid Nanoparticle Size and Polydispersity by Dynamic Light Scattering

The determination of SPIKEVAX DP lipid nanoparticle (LNP) size and polydispersity index (PDI) was performed on January 14, 2022 per SOP-0998 "Determination of Particle Size Distribution and Polydispersity by Dynamic Light Scattering", submitted by the firm. Measurements were made using a (b) (4)

The test results obtained by (b) (6) and those from the firm are summarized in Table 2.

Table 2. Results of LNP size and PDI

Lot	CBER size (nm)	Firm size (nm)	CBER PDI	Firm PDI
031H21A	(b) (4) (b) (4) Average: (b) (4)	(b) (4)	(b) (4) (b) (4) Average: (b) (4)	(b) (4)
064H21A	(b) (4) (b) (4) Average: (b) (4)	(b) (4)	(b) (4) (b) (4) Average: (b) (4)	(b) (4)
940875	(b) (4) (b) (4) Average: (b) (4)	(b) (4)	(b) (4) (b) (4) Average: (b) (4)	(b) (4)

CBER test results from three DP samples lots were: (b) (4) nm for particle sizes, (b) (4) for PDI. The results all met the release specifications: (b) (4) nm for LNP (b) (4) and (b) (4) for PDI and were similar to the results from the firm.